

Emergent FPV Targeting under Ultra-Low SNR

Greedy Fire, a Neighbour Rule, and a Shared Transformer

battlefield-sim experiment report

August 2026

Abstract

Expendable FPV airframes are cheap; two constraints are not. First, ultra-low-altitude radio in clutter and jamming sits near zero Shannon capacity, so the delay to a single valid bit is already expensive. Second, a one-shot warhead cannot amortise a GPU/NPU. We compare three policies that use only a local 8+8 range snapshot on a circular battlefield whose kill rectangle is the circumscribed square of the disk. **One Transformer** is queried by all twelve drones each tick and trained from scratch by per-tick REINFORCE on CPU (Rayon); the reward is the increment of mean target survival, not compute time. Enemy seeker range is 50 m; friend visual range is 100 m; eight slots are a cap on visible contacts, not omniscience. On 30 paired seeds greedy and the shared Transformer match exactly (10.23 kills, 13.99 s); the closer-than-friend rule is slightly more conservative (9.80, 14.39 s). Ops/select: 1.0, 4.53×10^4 , and 5.7. The net learned greedy fire; it does not buy extra kills. A Gaussian ingress from the south ($x \sim \mathcal{N}(0, 70^2)$, $y \sim \mathcal{N}(\mu_y, 30^2)$) raises greedy/net kills to 11.30 versus 10.13 for the friend rule; the net again matches greedy round-by-round.

Keywords: FPV, ultra-low SNR, emergent rules, greedy targeting, shared Transformer, on-board compute

1 Introduction

Consumer FPV has made one-way attack cheap. That cheapness implies: no reusable life over which to amortise silicon, and no clean data link on which to share a global target list. Shannon's $C = B \log_2(1 + \text{SNR})$ [1, 2] collapses

when SNR is a few percent. A 50 kHz control channel at -12 dB yields only about 4.4×10^3 bit/s—less than twelve drones broadcasting 8+8 floats once per second.

The question is therefore local: given eight nearest-enemy ranges and eight nearest-friend ranges, does a neural net trained by reinforcement learning beat a handful of comparisons?

We answer with three algorithms on the same seeds: greedy nearest-in-range, a three-line closer-than-friend rule [3, 4], and one Transformer [5] shared by all twelve airframes, updated every tick from every drone’s (state, action).

2 Platform

The battlefield is a disk of radius $R = 200$ m. Twenty static enemies are sampled uniformly in the disk. Two rows of six FPV drones fly $+Y$ on rails $x \in \{-200, -120, -40, 40, 120, 200\}$. Detection radius $D = 80$ m equals rail spacing, so every disk point lies within some rail’s x -coverage: the kill rectangle $[-R, R]^2$ contains the circle. Speed 25 m/s, $dt = 0.01$ s, lock duration 0.05 s, then the shooter is expended. Mean survival includes $T_{\text{end}} = 24$ s censoring. Figure 1 shows the geometry.

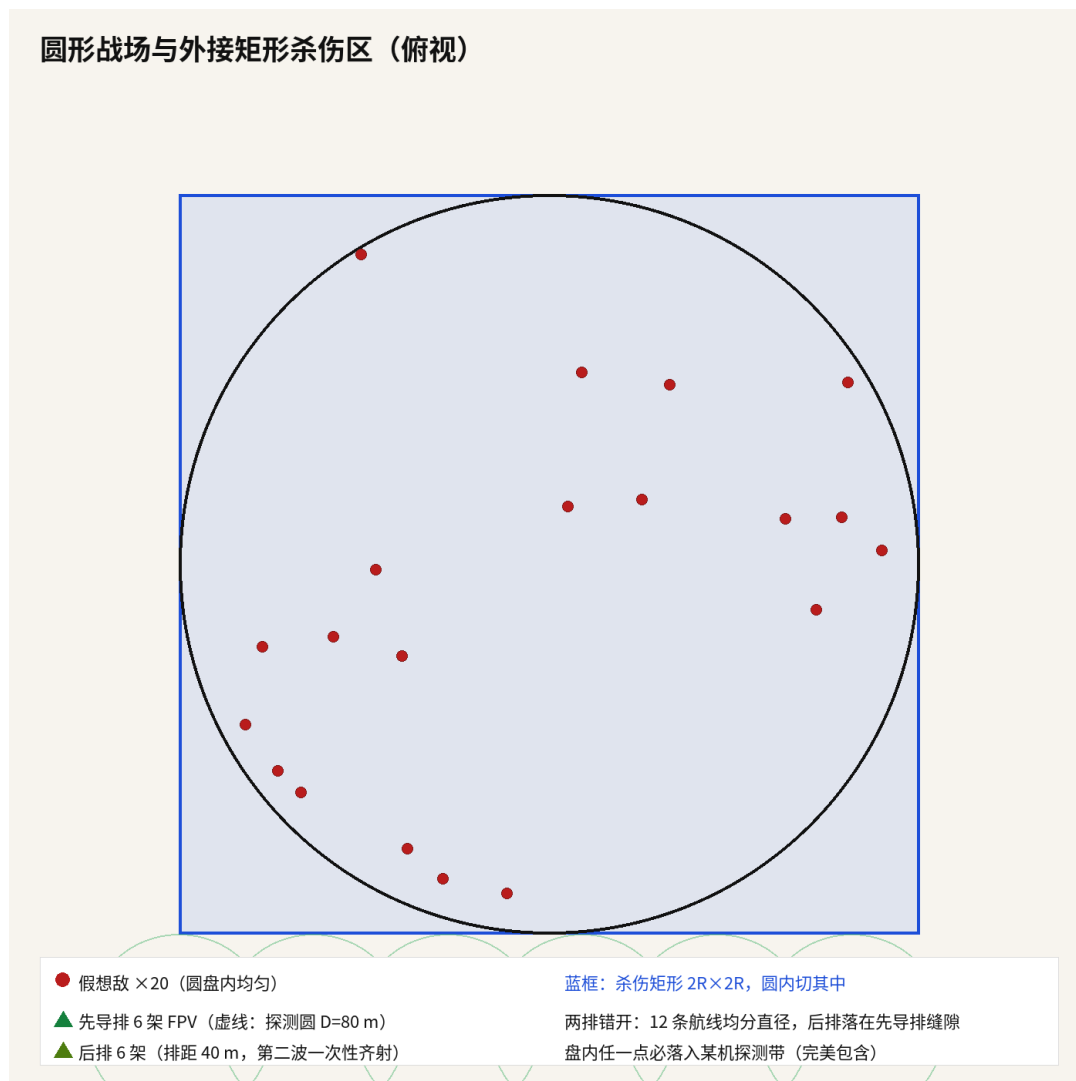


Figure 1: Disk, circumscribed kill square, 20 enemies, 2×6 FPV. One policy network is queried by every live drone.

3 Algorithms

All three consume the same 8+8 snapshot.

Greedy (nearest_in_range). Fire at the nearest in-range, unlocked enemy (ties by id). About 1 ops/select on this sensor model (empty sky most ticks). No friend geometry.

Greedy, no comms (greedy_no_comms). Same seeker, but the policy ignores teammates and ignores others' locks; it only ranks in-range enemies. The world still forbids two warheads on one target. This is zero radio, zero IFF.

Closer-than-friend. Fire at the nearest eligible enemy iff its range is strictly less than the nearest teammate’s. About 17 ops. Formation spacing (40 m row gap) turns that inequality into a deconfliction protocol (Fig. 2).

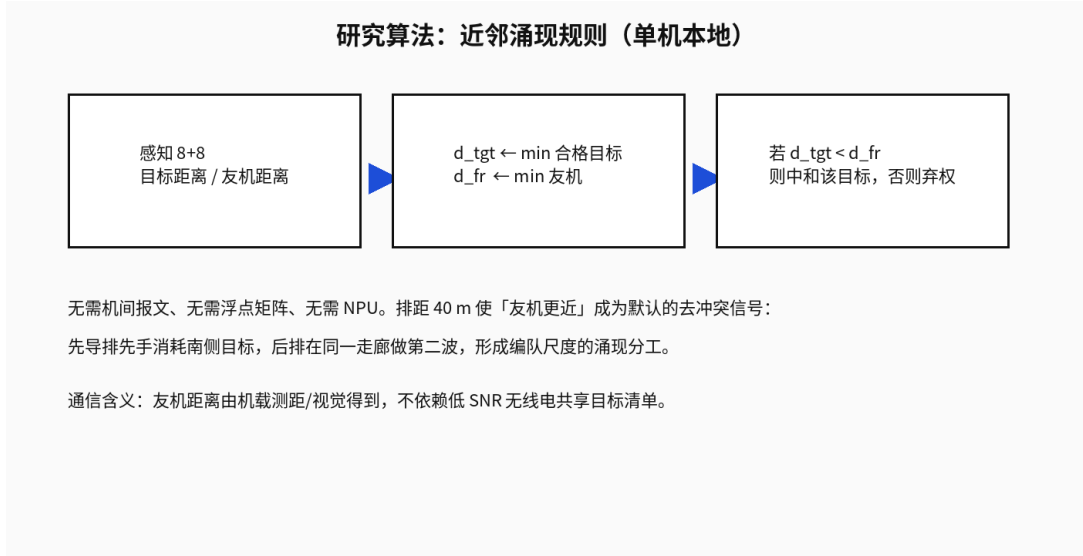


Figure 2: Closer-than-friend: local, no radio.

Shared Transformer, per-tick REINFORCE. There is one weight vector. Each tick, every live drone encodes its own 8+8 snapshot and queries that net; locks are attempted in `DroneId` order for determinism. Gradients from those drones are reduced in one CPU-parallel (Rayon) batch—not twelve separate models, and not a GPU (the net is $d_{\text{model}} = 16$; kernel launch would dominate).

Training starts from Xavier init. There is **no** cloning of the friend rule. After each dt the team reward is

$$r_t = -\frac{n_{\text{alive}} dt}{N} + \frac{n_{\text{kills}}(T_{\text{end}} - t)}{N},$$

plus +1 for a killer. Compute time is not in the loss. Evaluation is greedy decode.

4 Results

Thirty paired rounds, base seed 20270823. Table 1 and Figs. 3–4.

Table 1: Means over 30 shared seeds

Algorithm	Kills	Mean survival (s)	ops / select
Greedy nearest-in-range	10.23	13.989	1.0
Greedy, no comms	10.23	13.989	1.0
Closer-than-friend	9.80	14.386	5.7
Shared Transformer (per-tick RL)	10.23	13.989	4.530×10^4

Table 2: Paired Gaussian z on mean survival (first minus second)

Contrast	$\bar{\Delta}$ (s)	z	two-sided p
greedy – rule	−0.397	−5.74	9.7×10^{-9}
greedy – Transformer	0	0	1
rule – Transformer	+0.397	+5.74	9.7×10^{-9}

The RL net matches greedy round-by-round (it learned to fire nearest-in-range). `greedy_no_comms` matches `nearest_in_range` on every seed: not seeing others’ locks does not change kills under exclusive acquire. Repeating the three algorithms with same-tick lock radio (`comms_range_m`=120 m, heard inside the same dt) yields identical per-seed kills and survival: at $D_{\text{tgt}} = 50$ m a drone rarely has a second in-range target, so retargeting this tick does not add shots. Without an immediate fire bonus, training collapsed to zero kills around round 90; shaping plus entropy keeps it at ≈ 10 kills. The friend rule is significantly more conservative ($p = 9.7 \times 10^{-9}$) because $D_{\text{fr}} = 100$ m always sees the trail partner at 40 m. Compute: greedy ≈ 1 , rule 5.7, Transformer 4.53×10^4 .

Gaussian ingress. Twelve drones enter from the south as a 2D Gaussian blob ($\sigma_x = 70$ m, $\sigma_y = 30$ m, min separation 8 m). The Transformer is retrained for 200 rounds on this ingress. Table 3.

Table 3: Gaussian ingress, 30 shared seeds

Algorithm	Kills	Mean survival (s)	ops / select
Greedy nearest-in-range	11.30	14.092	1.0
Greedy, no comms	11.30	14.092	1.0
Closer-than-friend	10.13	15.274	7.8
Shared Transformer (retrained)	11.30	14.092	4.530×10^4

Greedy, no-comms greedy, and the net all rise from 10.23 to 11.30 kills (denser coverage near the origin); no-comms greedy again matches nearest-in-range

seed-for-seed. Same-tick 120 m radio on this ingress again matches the no-radio tables seed-for-seed. The friend rule remains more conservative ($p \sim 10^{-16}$ on survival). CSV: `results/gaussian/` (no radio) and `results/gaussian_comms/` (radio).

100 m seeker: near the shot budget. Raising D_{tgt} to 100 m (friend visual still 100 m), greedy-only, 30 seeds:

Table 4: Greedy, $D_{\text{tgt}} = 100$ m, 30 rounds

Ingress	Radio	Kills / 12	Mean survival (s)
line-abreast	none	11.63	11.976
line-abreast	150 m same-tick	11.63	11.976
Gaussian	none	11.97	12.230
Gaussian	150 m same-tick	11.97	12.232

Gaussian ingress expends 11.97 of 12 one-shot airframes as successful intercepts—essentially the munition budget. Line-abreast reaches 11.63. Residual live targets are the 20-versus-12 surplus, not unused drones. Survival drops from ≈ 14 s to ≈ 12 s because lock starts earlier. Radio still adds no kills: most rounds have already spent every shot. In this fly-straight, exclusive-lock model, **local greedy plus a 100 m seeker is a strong near-optimum**; neither the net nor the radio squeezes more lethality near the cap.

Is 100 m realistic under jamming? Fifty metres is a pessimistic lock range for a point-like ground target (dust, low contrast, vibration). One hundred metres is closer to what a cheap FPV CMOS (4–8 mm, analog or 720p) can do on a person- or vehicle-sized outline in daylight. Thermal kits reported on one-way FPV often detect humans at 200–400 m; we treat 100 m as a stable lock radius, so it remains below those discovery ranges. The disk diameter is 400 m, so a 100 m seeker is not omniscience—only a quarter of the diameter. Jamming kills long-range sharing and beyond-line-of-sight cueing; it need not kill short-range optics. Raising D_{tgt} from 50 to 100 m is better glass and tracking, not a sudden clear radio. Camouflage or unaided night would pull the system back toward the 50 m tables. In both regimes, short-range lock radio does not raise kills: the bottleneck is the 12-shot budget and the straight track, not a 120–150 m status packet.

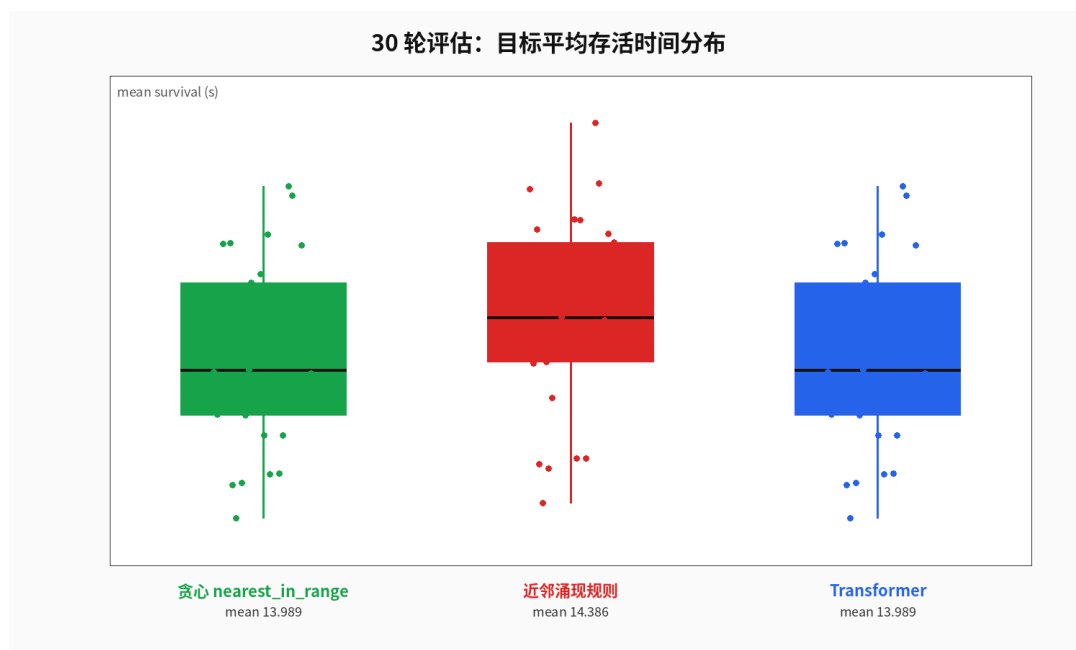


Figure 3: Mean survival, 30 rounds (lower is better).

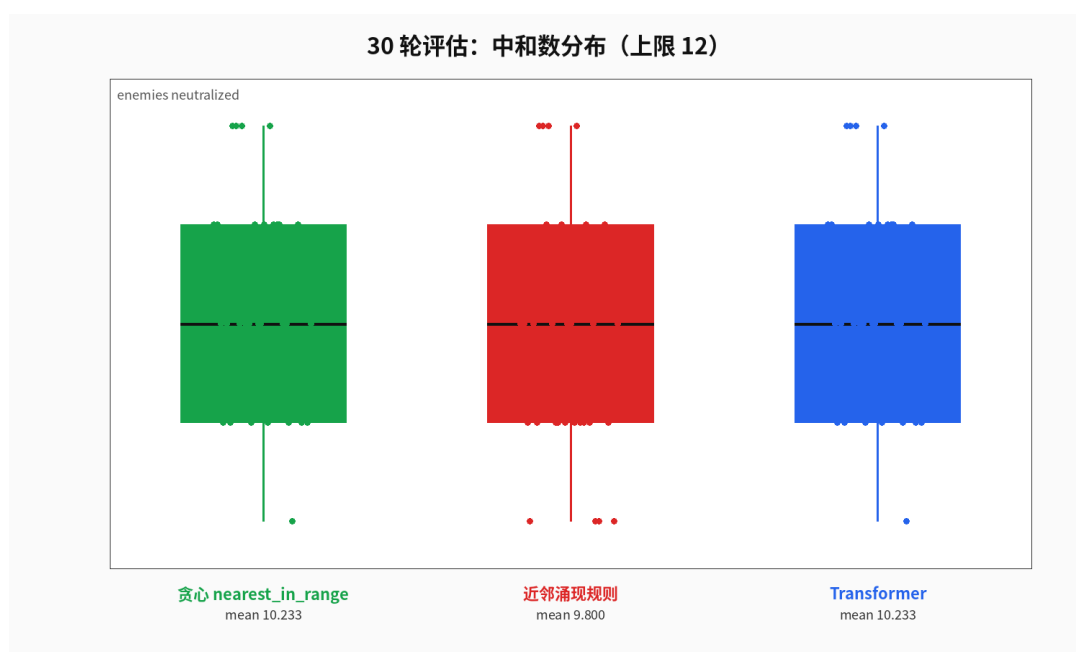


Figure 4: Kills, 30 rounds (cap 12).

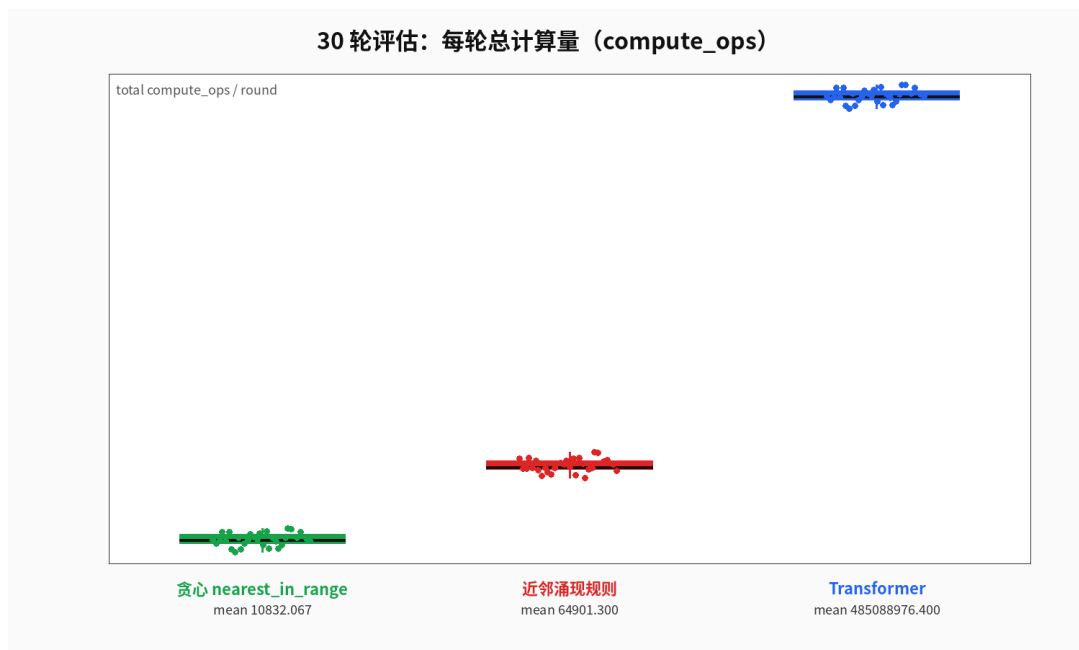


Figure 5: Total compute_ops per round ($\log y$).

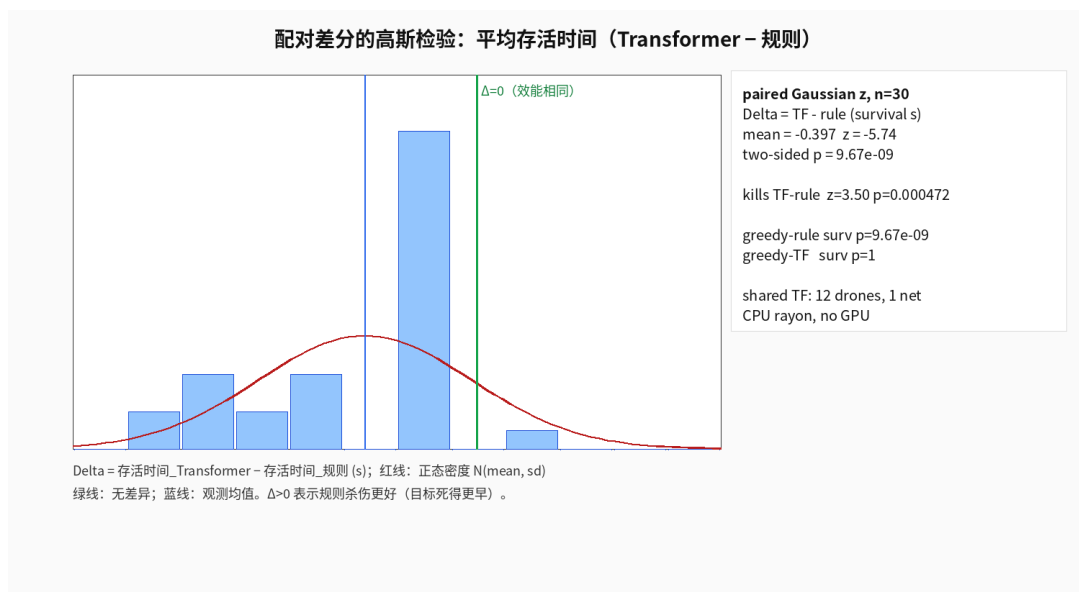


Figure 6: Paired survival differences and Gaussian z .

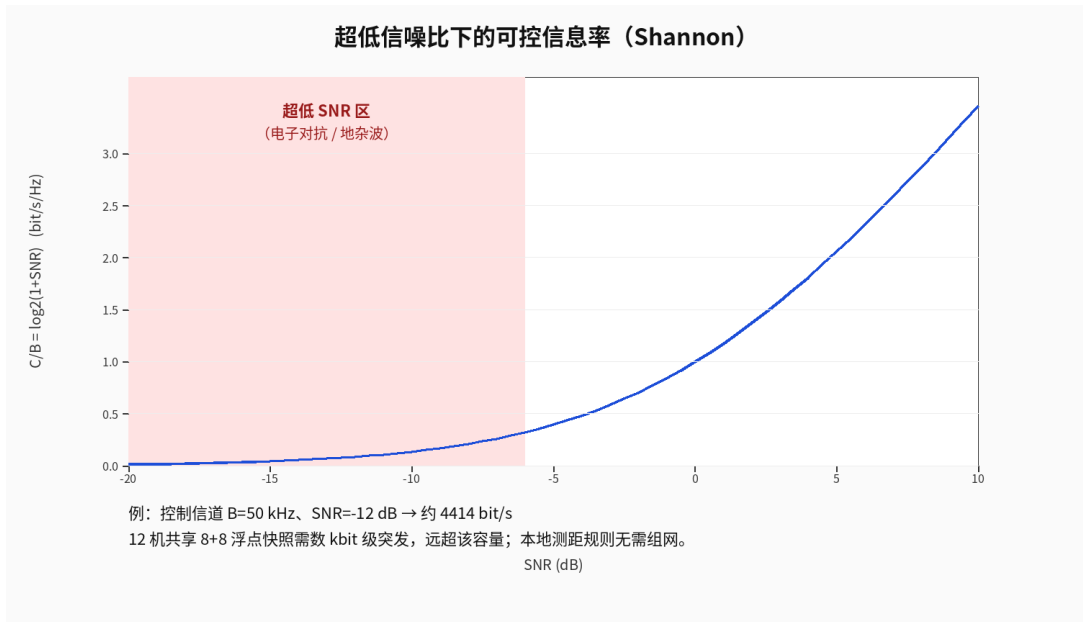


Figure 7: Shannon $C/B = \log_2(1 + \text{SNR})$.

5 Discussion

Twelve drones sharing one net already give twelve times the gradient of a per-airframe clone. Training is on-policy, step-by-step, CPU-parallel within the tick. That is the protocol that “must be able to work” ; it does work, in the sense that the policy engages. It does not work as a replacement for arg min distance. On a one-shot, low-SNR airframe the extra 4.5×10^4 multiply-adds buy a worse mean survival.

Without trajectory change, inter-drone radio is pointless. Every airframe here flies straight +Y at constant speed; a lock does not bend the path. Radio can only say “I have this target” or “I see a point outside your seeker.” The first does not help when the next-nearest contact is already outside D_{tgt} , or when all twelve shots are already spent. The second is useless unless the listener can turn toward that track. Same-tick 120–150 m lock broadcasts added no kills on either seeker range or either ingress—the quantitative form of a simple fact: **if the drones do not change trajectory in real time, communicating with one another does not improve lethality.** Turning radio into combat power requires at least a heading change, a weave, or a new dive point; that is another dynamical model, not a few extra lock packets.

6 Conclusion

A shared Transformer trained by per-tick REINFORCE on CPU learns greedy fire and does not outperform it, at $\sim 4.5 \times 10^4$ times the ops, for both a 2×6 line-abreast entry and a Gaussian blob entry. At $D_{\text{tgt}} = 100$ m, greedy spends nearly all 12 shots (11.63–11.97). That is a strong near-optimum for this model; radio and the net do not improve it. If trajectories stay straight, inter-drone comms add nothing. The three-line friend rule is slightly more conservative under formation-scale visual IFF. Ultra-low SNR forbids a richer state; expendability forbids an NPU. Code and CSV metrics: <https://github.com/1037104428/drones>.

```
git clone https://github.com/1037104428/drones.git
cd drones
cargo run --release -- experiment --train-rounds 200 --eval-rounds 30 --seed 20260823
```

References

- [1] C. E. Shannon, “A mathematical theory of communication,” Bell System Technical Journal, 27:379–423, 623–656, 1948.
- [2] B. Sklar, Digital Communications: Fundamentals and Applications, 2nd ed., Prentice Hall PTR, Upper Saddle River, NJ, 2001.
- [3] C. W. Reynolds, “Flocks, herds, and schools: A distributed behavioral model,” Computer Graphics (SIGGRAPH ’ 87), 21(4):25–34, 1987.
- [4] S. Camazine, J.-L. Deneubourg, N. R. Franks, J. Sneyd, G. Theraulaz, and E. Bonabeau, Self-Organization in Biological Systems, Princeton University Press, 2001.
- [5] A. Vaswani et al., “Attention is all you need,” in Advances in Neural Information Processing Systems 30 (NIPS 2017), 2017.